Formal Epistemic Structures and Mechanics — V4 science update

Paper VI current science manuscript overlay
Date: 2026-08-19
Base manuscripts: `FINAL_V2_1.md` + `FINAL_V3.md` (historical scientific layers retained)
Current successor evidence: `research/claim_expansion/p6/P6_X2_*`
Science terminal:
`P6_CERTIFICATE_LIFTING_SEMANTICS_SUPPORTED__BOUNDED_FORMAL_DONOR_STACK__IDEAL_PRODUCT_EQUIVALENT`

V4 keeps all V2.1/V3 theorems and changes the top-level interpretation from a primarily negative erasure/separation result to a constructive certificate-lifting architecture.

Replacement abstract for V4

Autonomous scientific systems increasingly operate inside mature governance and reproducibility machinery: runtime proof-of-execution, certified proposal–execution traces, portable action/approval receipts, workflow provenance signatures, dependency/effect systems, and attested execution boundaries. P6 treats these mechanisms as reusable donor certificates rather than competitors. The question is no longer merely whether computation can remain valid while scientific admissibility changes; it is how strong lower-level certificates can be composed, preserved and selectively lifted into continued scientific standing when a dynamic scientific state changes.

We introduce a bounded certificate-lifting semantics. A valid donor certificate remains valid in its native theory. Scientific preservation is a separate lift requiring claim/content binding, measurement semantics, evidence semantics, inferential obligations and scientific epoch to remain preserved or be revalidated. Multiple valid donor certificates cannot manufacture a missing scientific continuity coordinate merely by accumulation. Conversely, a change need not invalidate the whole certificate stack: only scientifically affected coordinates require revalidation, while unaffected execution, replay, provenance, workflow and attestation certificates remain reusable. An ideal donor product enriched with the same scientific fields and lifting rule is extensionally equivalent, ruling out any inherent centralization advantage.

The exhaustive finite model evaluates 320 donor-certificate/scientific-extension states, with zero donor-conservativity violations and zero ideal-product mismatches. It contains 25 minimal one-coordinate lifting separations, 31 certificate-product non-laundering countermodels, 155 exact full-revalidation successes and 1,055 failures of proper-subset revalidation. An independent implementation reproduces the canonical enumeration. The contribution is therefore a conservative scientific certificate-lifting and selective-revalidation calculus over absorbed certificate systems, not generic certification, provenance, authorization, execution attestation or deployed-agent superiority.

New donor-engulfment interpretation

V4 explicitly adopts the strongest useful structure of modern certificate systems.

- **Proof-of-execution systems** contribute contract-bound authorization, path compliance, history integrity and replayable execution attestation.
- **Certified-trace / proposal-certification-execution systems** contribute a hard separation between generation, certification and effectful execution.
- **Proof-carrying action systems** contribute runtime-neutral action identity, approval binding, runtime/outcome receipts and portability.
- **Scientific workflow signature systems** contribute formal reproducibility tenets, workflow/data provenance and cryptographic execution signatures.
- **Certified-purity systems** contribute structural exclusion of undeclared effects, signed purity certificates and attestation.

P6 claims none of those atoms. Their strongest certificates are admitted directly into the P6 certificate stack.

The improvement is the scientific **lift interface**: what additional claim-specific continuity evidence allows a still-valid lower-level certificate to support preservation of scientific standing after change, and what can be retained when the lift fails.

18. Certificate lifting over donor-complete execution and workflow certificates

Let `d` be a donor certificate validated by its native theory and let `s=(c,m,e,i,t)` represent, in the bounded V4 instance, claim/content binding, measurement semantics, evidence semantics, inferential obligation and scientific epoch.

Define

```
Liftable(d,s) := DonorValid(d) AND c AND m AND e AND i AND t.
```

The five coordinates are a registered theorem instance, not a universal ontology. A donor that already carries an equivalent coordinate and rule is enriched rather than treated as deficient.

Theorem V4.1 — donor conservativity

Projection from the lifted semantics to the donor certificate preserves the donor-native validity judgment. P6 does not make a valid proof-of-execution, action, workflow or attestation certificate invalid merely to create a scientific distinction.

Theorem V4.2 — lifting separation

For every registered donor-certificate family and every non-inert scientific lift coordinate, there are two extensions of the same valid donor certificate that differ only on that coordinate and have different scientific-lifting judgments.

Thus the validity of a runtime/workflow/action certificate can survive a material scientific change even when the old scientific standing cannot.

Theorem V4.3 — certificate-product non-laundering

A product of native-valid donor certificates does not infer an absent scientific lift coordinate without an explicit bridge rule. Accumulating execution, replay, provenance, workflow and attestation certificates is not itself evidence that a claim-specific measurement or inference obligation remains discharged.

Theorem V4.4 — exact selective revalidation

Suppose a transition changes a nonempty set `S` of load-bearing scientific lift coordinates while the donor certificate remains valid. Revalidating all coordinates in `S` restores the lift when unchanged coordinates remain valid; revalidating any proper subset of `S` does not.

This turns the V3 separation result into a positive preservation rule: do not discard certificates whose native subject did not change; revalidate exactly the affected scientific bridge.

Theorem V4.5 — ideal-product equivalence

An information-equivalent donor product carrying the exact same scientific coordinates and lift predicate agrees extensionally with P6. The architecture claim is therefore about a reusable scientific lifting interface, not centralized expressive power.

Exhaustive finite support

Registered donor-certificate families: proof-of-execution, certified trace/PCE, proof-carrying action, workflow execution signature, and certified purity/attestation.

Registered scientific lift coordinates: claim/content binding, measurement semantics, evidence semantics, inferential obligation, scientific epoch.

Exact enumeration: - states: **320**; - donor-conservativity violations: **0**; - minimal one-coordinate separation witnesses: **25**; - all-donor-product scientific non-laundering countermodels: **31**; - exact full-revalidation successes: **155**; - proper-subset revalidation failures: **1,055**; - ideal enriched-product mismatches: **0**; - canonical row SHA-256: `e1e3c48bcefea3750d952c6b0ff37ac660a2e21f9823fdfdeb50bb62e819ff93`.

The independent checker reconstructs the enumeration without importing the primary checker.

Wider P6 claim

P6 now supports the following bounded general statement:

> Strong execution, action, workflow and attestation certificates can be composed and preserved as lower-level objects in dynamic scientific computation, but preservation of scientific standing requires an explicit lift across claim-specific semantic obligations; scientifically material changes trigger exact revalidation of affected lift coordinates without discarding unrelated valid donor certificates.

This is wider than the V3 erasure claim because it says how the strongest donor systems are **absorbed, composed and improved**, not merely why a weaker projection can lose information.

Limits

V4 does not claim the five registered lift coordinates are universally minimal, that current certificate systems fail their native goals, that ORION is inherently more expressive than an equally typed product, or that the formal result establishes deployed-agent utility. A future donor carrying an equivalent scientific lift relation should be absorbed and would move that embedding to the equivalence side of the theorem.

Replacement conclusion for V4

The strongest version of P6 is not a rejection of existing certificate systems. It is a way to use them more completely. Runtime proof, certified traces, portable action receipts, workflow signatures, provenance, dependency/effect semantics and execution attestation remain valuable and reusable after they are absorbed into the scientific stack. What ORION adds is a typed bridge from those operational certificates to preservation of scientific standing, plus a rule for selective revalidation when the bridge is disrupted.

This yields a broader architectural result: scientific computation can preserve mature lower-level certificates across change rather than rebuilding trust from scratch, while refusing to equate lower-level validity with unchanged scientific entitlement. The ideal enriched product equivalence keeps the claim honest: the contribution is the explicit scientific lifting semantics and its composition/revalidation laws, not centralization.

Current science terminal:
`P6_CERTIFICATE_LIFTING_SEMANTICS_SUPPORTED__BOUNDED_FORMAL_DONOR_STACK__IDEAL_PRODUCT_EQUIVALENT.`

---

source: papers/paper-06-formal-epistemic-structures-and-mechanics/manuscript/FINAL_V4.md
sha256: 74031304eee609c06aefd16f3a3a7febda1925db70ffde936489303d680ddb3d
commit: 37e82ceb54afc12b1d9958895d3d110b2b567858
rendered: 2026-08-22T00:09:35+00:00 by scripts/build_paper_pdf.py (WeasyPrint; not the LaTeX tree)